

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination 2022

Course: B. Tech. Branch: Electrical Engineering

Semester :7th

Subject Code & Name: BTEEC 703/ Electrical Drives

Max Marks: 60

Date: 01-02-2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) What is an Electrical Drive system? Explain Basic structure of electrical drives.	Understand	6
B) Derive the fundamental torque equation of electric drive.	Evaluate	6
C) Explain Multi-Quadrant operation of Electrical Drive with suitable example.	Remember	6
Q.2 Solve Any Two of the following.		12
A) Explain the working of close loop speed control scheme of electric drive with inner current loop.	Understand	6
B) Discuss the various modes of operation of electric drives along with drive classification.	Remember	6
C) Explain in detail the "Classes of motor duty".	Knowledge	6
Q.3 Solve Any Two of the following.		12
A) Explain the working of dual converter fed DC drive system.	Understand	6
B) Explain the working of single-phase full converter fed DC motor drive. Draw the speed torque characteristic for different firing angles.	Evaluate	6
C) Explain basic characteristics of DC Motor.	Remember	6
Q.4 Solve Any Two of the following.		12
A) How does a "Slip power recovery scheme" works in three phase Induction motor.	Knowledge	6
B) Discuss the multi-quadrant operation of Induction motor drives fed from voltage source inverters.	Understand	6
C) Why starters are used in three phase Induction motors? Explain different type of starters used in three phase Induction motors.	Knowledge	6

Q. 5 Solve Any Two of the following.		12
A) Explain the working of load commutated inverter fed synchronous motor drive in detail.	Remember	6
B) Discuss the close loop control of synchronous motor drive.	Understand	6
C) Amongst the following industries explain the working of drives used for specific applications for any one industry in detail:	Knowledge	6
(i) Textile mill		
(ii) Steel Rolling mill		
(iii) Cement mill		
(iv) Sugar mill		

***** End *****